

How these sheets work

One sheet per competency, 6 sheets for this video. Each gives you two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before the video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing, some a grid, some three panels. The division is yours.
4. Words go inside the drawing, as labels and inside little boxes with arrows. A sentence running the width of the box does not count.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a grid or a sequence, so whichever way you think, there is a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set it up on a tablet, and write the prompt letter you chose at the top. Hand ruled counts exactly the same.

This is anatomy

Every prompt asks what a structure is, where it sits, and what it is next to. Function belongs to physiology and is not on these sheets.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

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WEEK 1 · TISSUES AND HISTOLOGY

Embryonic germ layers

COMP

You should be able to: Name the three primary germ layers, ectoderm, mesoderm, and endoderm, and the tissues each one gives rise to.

PICK ONE AND DO THAT ONE

- ☐ **A THREE-LAYER EMBRYO**
Draw an embryo as three stacked layers, ectoderm on top, mesoderm in the middle, endoderm below. From each layer, draw arrows out to small drawings of the tissues it becomes, and label them.

Then: Mark which layer gives all four tissue types something, and which layer gives only one.
- ☐ **B THREE-COLUMN TABLE**
Draw three columns headed ectoderm, mesoderm, endoderm. Under each, draw the tissues and organs it gives rise to, and label them.

Then: Circle the tissue type that comes from all three layers.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

split it up however the prompt needs

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COMP

WEEK 1 · TISSUES AND HISTOLOGY

Epithelial tissue identification

You should be able to: Classify epithelial tissue by cell layers and cell shape and identify the basement membrane, apical specializations, and glandular epithelium on a slide.

PICK ONE AND DO THAT ONE

☐ **A SLIDE GALLERY**

Draw eight small slide views: simple squamous, simple cuboidal, simple columnar, stratified squamous (keratinized and not), stratified cuboidal, pseudostratified ciliated columnar, transitional. In each, draw the basement membrane, the free surface, and the nuclei where they really sit. Label each one.

Then: Add microvilli, cilia, or goblet cells to the slides that have them.

☐ **B LAYERS BY SHAPE GRID**

Draw a grid with simple and stratified down the side and squamous, cuboidal, columnar across the top. In each cell, draw the tissue and write one place in the body it lines. Add a row for the two that break the rule, pseudostratified and transitional.

Then: Draw a gland beside the grid and label the epithelium it is made from.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

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Dr. Sharilyn Rennie

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COMP

WEEK 2 · TISSUES AND HISTOLOGY

Connective tissue identification

You should be able to: Describe connective tissue as cells, fibers, and ground substance and recognize the connective tissue proper subtypes, cartilage, bone, and blood on a slide.

PICK ONE AND DO THAT ONE

☐ **A SLIDE GALLERY**

Draw small slide views of areolar, adipose, reticular, dense regular, dense irregular, elastic, hyaline cartilage, elastic cartilage, fibrocartilage, bone, and blood. In each, draw the cells, the fibers, and the ground substance so the three parts are visible. Label each tissue.

Then: Circle the one feature on each slide that tells you which tissue it is.

☐ **B FAMILY TREE**

Draw a tree that starts at connective tissue and branches into connective tissue proper, cartilage, bone, and blood, then branches again to every subtype. At the end of each branch, draw the tissue small and name one place it lives.

Then: Write the three parts every connective tissue has at the top of the tree, with a sketch of each.

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WEEK 3 · TISSUES AND HISTOLOGY

COMP

**Muscle tissue
identification**

You should be able to: Recognize the three muscle tissue types, skeletal, cardiac, and smooth, by striations, cell shape, and number of nuclei.

PICK ONE AND DO THAT ONE

☐ **A THREE SLIDES SIDE BY SIDE**

Draw skeletal, cardiac, and smooth muscle as three slide views side by side. Draw the striations where they exist, the cell shape, the nuclei in the right number and position, and the intercalated discs in cardiac. Label each.

Then: Under each, write one organ where you would find it.

☐ **B TELL-THEM-APART TABLE**

Draw a table with the three muscle types down the side. Across the top: striations, cell shape, number and position of nuclei, one special feature, and a tiny sketch.

Then: Circle the one cell in the table that lets you tell cardiac from skeletal at a glance.

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PASS 2 COLOR _____

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COMP

WEEK 3 · TISSUES AND HISTOLOGY

Nervous tissue identification

You should be able to: Identify neurons and neuroglia in nervous tissue and the parts of a multipolar neuron.

PICK ONE AND DO THAT ONE

☐ **A LABELED NEURON ON A SLIDE**

Draw a multipolar neuron large as it looks on the slide: cell body, nucleus, nucleolus, dendrites, axon hillock, axon, axon terminals. Draw several small neuroglia around it and label them as glia.

Then: Mark which direction a signal travels along your drawing.

☐ **B TWO-BOX COMPARISON**

Draw a line down the middle. Left, a neuron with every part labeled. Right, neuroglia drawn small and numerous. Under each, write what you look for on the slide to tell them apart.

Then: Draw the one part of the neuron that can be very long, and mark how long it can get relative to the cell body.

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WEEK 3 · TISSUES AND HISTOLOGY

Body membranes

COMP

You should be able to: Distinguish the cutaneous, mucous, serous, and synovial membranes by composition and location.

PICK ONE AND DO THAT ONE

☐ **A** FOUR MEMBRANES IN SECTION

Draw the four membranes as four small cross sections: cutaneous, mucous, serous, synovial. In each, draw the epithelium if there is one and the connective tissue under it. Label the layers and write where in the body it is found.

Then: Mark the one membrane that has no epithelium.

☐ **B** SORTING TABLE

Draw a table with the four membranes down the side. Across the top: the epithelium, the connective tissue, whether it faces the outside world or a closed space, and one location. Draw the layers in the first two columns.

Then: Circle the membrane that lines a cavity that opens to the outside.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

split it up however the prompt needs